TED (15) – 3132 (REVISION – 2015)

Model Question Paper

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

DATABASE MANAGEMENT SYSTEMS

Time: 3 hours

(Maximum marks : 100)

PART – A

Marks

I. Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Distinguish between data and information
2. List the various database users
3. Define composite key.
4. Give an example for creating index.
5. State the technology of datamining.

(5x2=10)

PART - B

II. Answer any five of the following. Each question carries 6 marks.

1. Discuss the component modules of DBMS with a neat diagram
2. Explain the various notations used in Entiity-Relationship diagram.
3. Illustrate the technology of mapping ER model to relational model.
4. Create a table of students and perform the following operations using SQL.
   (i) Enter the rollno, name, age, city and grade of 5 students.
   (ii) Find the city of students who got grade 'A';
5. Create a view of a company with the following entities
   (i) employee(firstname,lastname,birthdate,address,ssn, sex,salary,deptnumber)
   (ii) department (deptnumber,deptname)
   (ii) project(projectno, projectname,projectlocation).
   (Use appropriate abbreviations for the attributes as required)
6. Differentiate parallel DBMS and distributed DBMS.

7.Explain the steps taken while writing java application program with database access through JDBC function calls.

(5x6=30)

PART – C

(Answer one full question from each unit. Each question carries 15 marks.)

Unit – I

III. (a) List and explain the applications of DBMS (10)

(b) Differentiate between DDL and DML (5)

OR

IV. (a) Compare hierarchical, network and relational model (9)

(b) Write notes on Centralized and Client/Server Architecture (6)

Unit – II

V. (a) Explain E-R model with a suitable example (8)

(b) Discuss the various keys in RDBMS (7)

OR

VI. (a) Explain enhanced ER diagram. Draw an example of ER diagram of an employee showing sub class and super class (8)

(b) Describe the different types of join operations (7)

Unit – III

VII. (a) Ilustrate the method of adding NULL,CHECK,PRIMARY KEY and FOREIGN KEY constraints with examples. (8)

(b) Explain the different transaction states with help of diagram (7)

OR

VIII. (a) Explain the method of authorizing on data and granting & revoking of privileges (9)

(b) Create a procedure to read a student's name and mark, and print his grade assuming (i) mark>=90 - 'A' (ii)mark >=80 -'B' (iii) mark>=70 - 'C' (iv) mark>=60 -'D' (v) mark>=50 -'E' else 'F' (6)

Unit – IV

IX. (a) Explain functional dependency (5)

(b)

(i) Define normalization (2)

(ii) Discuss the different types of normal forms (8)

OR

X. (a) Describe the different goals of data-mining and knowledge discovery (8)

(b) Explain the various data management issues in mobile databases (7)